

# Human Emotion Detection from Voice

## Project Overview

Human Emotion Detection from Voice is a machine learning and deep learning project that detects human emotions from speech audio. The system analyzes voice characteristics and predicts emotions using audio processing techniques.

The project extracts important voice features and classifies emotions such as Happy, Sad, Angry, Fearful, and Neutral.

## Features

- Detect human emotions from voice input
- Audio feature extraction using Librosa
- Voice preprocessing and analysis
- Emotion prediction using trained models
- Supports multiple emotions

## Technologies Used

- Python
- TensorFlow
- Librosa
- NumPy
- Pandas
- Scikit-learn
- Streamlit

## Dataset

This project uses the RAVDESS dataset for training and testing.

Supported emotions:

- Happy
- Sad
- Angry
- Fearful
- Calm
- Neutral
- Surprise
- Disgust

## Project Structure

```
emotion_detector/  
├── data/RAVDESS/  
├── models/  
├── app.py  
├── train.py  
├── utils.py  
└── requirements.txt
```

## Working Process

1. Load audio dataset
2. Preprocess audio files
3. Extract audio features
4. Train the emotion detection model
5. Save trained model
6. Predict emotion from voice input

## Installation

Clone the repository:

```
git clone https://github.com/yourusername/kammapolu-shashikala.git
```

Move to project directory:

```
cd emotion_detector
```

Install required packages:

```
pip install -r requirements.txt
```

## Run the Project

Train the model:

```
python train.py
```

Run the application:

```
streamlit run app.py
```

## Expected Output

Input:

Voice Audio File

Output:

Predicted Emotion: Happy / Sad / Angry / Neutral

## Conclusion

This project demonstrates how machine learning and audio signal processing techniques can be used to identify human emotions from voice data.